

Tomato Leaf Disease Detection with Precaution Using EfficientNet-B3

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Abstract—Tomato cultivation is crucial to global agriculture, yet it is highly vulnerable to a variety of diseases and pests that significantly affect both yield and quality. In particular, fungal infections, viral pathogens like Yellow Leaf Curl Virus, and bacterial diseases such as Late Blight, are common threats that impair production. Effective and timely detection of these diseases is essential to minimize crop loss and ensure higher yields. This research explores the application of the EfficientNet-B3 deep learning model for the classification and early detection of tomato leaf diseases. By employing state-of-the-art image processing techniques and deep learning architectures, this study aims to develop a robust detection system that identifies diseased tomato leaves with high accuracy. We used the Plant Village dataset for training and testing purposes, achieving an overall classification accuracy of 99%, surpassing several traditional machine learning models. In addition to detecting and classifying diseases, the proposed system provides precautionary tips to help farmers take timely actions and prevent further spread. This scalable and computationally efficient solution leverages FastAPI to ensure high performance, assisting in early disease detection and management for improved agricultural outcomes. Furthermore, this paper discusses the importance of dataset quality, compares the model's performance with other deep learning approaches, and highlights future avenues for enhancing classification systems in agricultural settings.

Keywords—Tomato Leaf Disease, EfficientNet-B3, Deep Learning, Image Classification, Plant Village Dataset, Early Disease Detection, Precaution Tips, Agriculture, Convolutional Neural Network (CNN), FastAPI

I. INTRODUCTION

Tomato cultivation plays a crucial role in global agriculture, yet it is highly vulnerable to a variety of diseases and pests that significantly affect both yield and quality. Fungal infections, viral pathogens like Yellow Leaf Curl Virus, and bacterial diseases such as Late Blight are common threats that impair production [7, 8]. Effective and timely detection of these diseases is essential to minimize crop loss and ensure higher yields.

In recent years, deep learning techniques have emerged as powerful tools for addressing complex problems in various domains, including agriculture [1, 2]. Convolutional Neural Networks (CNNs), in particular, have demonstrated remarkable success in image classification tasks, making them well-suited for plant disease detection [3, 4]. The ability of CNNs to automatically learn hierarchical features from raw image data has revolutionized the field of computer vision and its applications in agricultural science.

Among the various CNN architectures, the EfficientNet family has gained prominence due to its ability to balance accuracy and computational efficiency [9]. This study leverages the EfficientNet-B3 model, which offers an optimal trade-off between model size and performance, making it particularly suitable for the complex task of leaf disease classification. The model's efficiency is crucial in agricultural applications, where rapid and accurate disease detection can significantly impact crop management decisions [3, 6].

To enhance the model's performance and generalization capabilities, this research employs a range of data preprocessing and augmentation techniques [1, 4]. These include image resizing, normalization, and various transformations such as rotations, flips, and brightness adjustments. Such techniques are crucial for improving the model's ability to recognize diseases under varying real-world conditions, addressing the challenge of translating laboratory results to practical field applications [10].

The Plant Village dataset serves as the foundation for training and testing the proposed model [5]. This comprehensive dataset encompasses a wide range of tomato leaf diseases, providing a robust basis for developing a highly accurate classification system. By fine-tuning the EfficientNet-B3 model on this dataset, this study aims to achieve state-of-the-art accuracy in distinguishing between various tomato leaf diseases while maintaining computational efficiency.

While previous studies have explored various CNN architectures for tomato disease classification, including VGG16, VGG19, and ResNet50 [4, 9], this research focuses on pushing the boundaries of classification accuracy and efficiency. The proposed approach aims to surpass the performance of traditional machine learning models and compete with the highest accuracies reported in recent literature, which range from 91.9% to 99.7% [4, 9].

A key innovation of this study is the integration of the trained model with FastAPI, a modern, high-performance web framework designed for building APIs with Python [8]. This integration enables real-time disease detection capabilities, allowing for rapid deployment of the model in practical agricultural settings. Additionally, the system provides precautionary tips alongside disease classification results, helping farmers take timely preventive actions to mitigate crop losses.

II. LITERATURE SURVEY

Recent years have seen significant advancements in the application of deep learning techniques for plant disease detection, particularly in tomato leaf disease classification. This expanded literature survey summarizes key contributions from at least 20 recent studies in this field:

Lawrence C. Ngugi et al. (2021) introduced KijaniNet, an innovative CNN model for semantic segmentation of tomato leaves and lesions, achieving mIoU scores of 0.8448 and 0.6257 for leaf and lesion pixel classes, respectively. Their study proposed a novel approach for disease recognition by cascading a segmentation network with a GoogleNet classifier [1]. K. Trivedi et al. (2021) applied CNN for tomato disease classification on 3000 images across nine diseases and healthy leaves, achieving an accuracy of 98.49% [2]. Yang Zhang et al. (2020) enhanced Faster RCNN with a residual depth network and optimized anchor boxes, resulting in a 2.71% improvement in recognition accuracy [3]. Lilian Mkonyi et al. (2020) evaluated CNN architectures, finding that VGG16 achieved 91.9% accuracy for tomato leaf classification, aiding early detection of Tuta Absoluta pest invasion [4].

Yashwant et al. (2021) combined the RANSAC algorithm with machine learning classifiers such as LR, MLP, and SVM, achieving 93.2% accuracy and 0.903 AUC on various plant datasets, including tomato [5]. Lawrence C. Ngugi et al. (2020) proposed a CNN-based segmentation network that processed 256x256 images swiftly on both GPU and CPU, achieving an accuracy of 96.5% [6]. Barburiceanu et al. (2021) introduced a hybrid approach combining CNN feature extraction with hand-crafted descriptors, achieving 97.25% accuracy [7]. Ahmed et al. (2021) developed a mobile app using CNN, trained on 97,216 images across 14 crops, achieving 94% classification accuracy across 38 diseases [8].

Lijuan et al. (2021) compared various machine learning and deep learning models on the Plant Village tomato dataset, with ResNet34 achieving top metrics of 99.7% accuracy and F1 score [9]. Natheer et al. (2022) developed a smartphone app using DenseNet for tomato disease classification without preprocessing, achieving 99% accuracy across ten categories [10]. Ahmed et al. (2022) proposed a lightweight deep neural architecture for tomato leaf disease classification, achieving 98.91% accuracy with reduced parameters [11]. Aversano et al. (2020) explored VGG with transfer learning, achieving 98.05% accuracy on tomato disease classification using a fine-tuned VGG16 model [12].

Ozbilge et al. (2022) created a compact CNN achieving 98.53% accuracy on the PlantVillage dataset, demonstrating the effectiveness of small architectures [13]. Karthik et al. (2020) introduced an attention-embedded residual CNN, yielding 98.42% accuracy in detecting tomato leaf diseases [14]. Brahimi et al. (2017) applied deep learning with AlexNet, achieving 99.18% accuracy on a dataset of 14,828 images [15]. Siti et al. (2020) utilized MobileNet V2 for tomato leaf disease classification, achieving 97.85% accuracy, demonstrating the suitability of lightweight architectures for mobile platforms [16].

Sembiring et al. (2020) developed a concise CNN achieving 97.69% accuracy, balancing performance and model size effectively [17]. Fuentes et al. (2017) designed a deep-learning-based detector for real-time recognition of tomato plant diseases and pests, achieving mAP scores of 83.06% for diseases and 96.25% for pests [18]. Chen et al. (2023) proposed LBFNet, a model using a three-channel attention mechanism and pruning, achieving 99.73% accuracy on the PlantVillage dataset [19].

Chen et al. (2020) combined ABCK-BWTR and B-ARNet for tomato leaf disease identification, achieving 98.96% accuracy through their hybrid approach [20].

These studies collectively demonstrate the rapid progress in applying deep learning to tomato leaf disease detection, with accuracies consistently above 97% and some approaches reaching over 99%. They highlight the importance of balancing model complexity with computational efficiency, a key consideration in the development of effective disease detection systems. The high accuracies achieved across various architectures underscore the potential of deep learning in revolutionizing agricultural practices and improving crop health management.

The implementation of EfficientNet-B3 in the current study builds upon these foundations, aiming to provide an optimal balance of accuracy, efficiency, and practical applicability in tomato leaf disease classification. With a reported accuracy of 99%, this approach is competitive with the state-of-the-art results in this field, while potentially offering advantages in terms of model efficiency and deployability.

The project contributes to this growing body of research by leveraging the EfficientNet-B3 architecture for tomato leaf disease classification. Through meticulous optimization and training procedures, the model achieves a remarkable 99% accuracy, positioning it at the forefront of current research in this domain.

Table I provides a concise comparison of different methodologies applied in plant disease recognition and their corresponding performance metrics.

TABLE I Model Accuracy Comparison

Methodology Used	Ref. No	Performance Metrics
CNN	[2]	98.49 %
Faster RCNN with ResNet-101	[3]	98.54%
CNN with VGG16, ResNet50, VGG19	[4]	91.9%
RANSAC Algorithm	[5]	93.2%
CNN with ResNet 34	[9]	99.2%
CNN with DenseNet 201	[10]	99%
Mobile Net V2	[16]	97.85%

III. PROPOSED METHODOLOGY

The proposed method for predicting leaf diseases integrates advanced deep learning techniques with an efficient model architecture to accurately classify leaf diseases based on image data. The system comprises three main stages: data preprocessing, model architecture and training, and real-time prediction using FastAPI for quick, scalable deployment. Additionally, it provides precautionary tips along with predictions to help farmers take timely preventive measures and reduce crop damage.

A. Data Preprocessing

Effective preprocessing is crucial for maximizing the performance of the classification model. In this work, images of diseased leaves are collected and standardized to a consistent input format. Each image is resized to a resolution of 256x256 pixels, ensuring that the input size remains uniform and manageable across the dataset. This standardization is necessary to adapt the images to the input requirements of the neural network while preserving critical visual features like textures and color variations, which are essential for disease identification.

The plant leaf disease images utilized in this study were obtained from the Plant Village dataset [5], which serves as a comprehensive resource for various plant health conditions. To diversify the dataset and improve the model's ability to generalize to unseen data, several data augmentation techniques are applied. Random rotations help the model handle leaf images captured from different angles, while horizontal and vertical flips introduce variability in the dataset, simulating changes in leaf orientation. Brightness and contrast adjustments are applied to enhance the model's robustness against varying lighting conditions, which is a common challenge in outdoor environments. After augmentation, the pixel values are normalized to lie within the range [0, 1]. Normalization ensures that all inputs are on a similar scale, improving the model's convergence speed during training. The normalization process follows the formula:

$$\frac{x' - \mu}{\sigma} \quad (1)$$

where x' represents the pixel value, μ is the mean, and σ is the standard deviation of pixel values across the dataset.

B. Model Architecture and Training

The architecture of the leaf disease classification model is based on EfficientNet-B3, a convolutional neural network (CNN) architecture optimized for both performance and parameter efficiency. EfficientNet's ability to balance accuracy and computational cost makes it ideal for real-world applications that need to process large amounts of data with limited resources, such as mobile devices or edge computing platforms. The model is pre-trained on ImageNet, which provides a robust set of image features related to object edges, textures, and patterns. This transfer learning approach allows the model to leverage prior knowledge about generic image features and focus more on the domain-specific task of leaf disease classification by fine-tuning the higher layers.

For the specific task of classifying leaf diseases, the final fully connected layers of EfficientNet are replaced with a custom classification head. This head consists of dense (fully connected) layers, which reduce the high-dimensional feature representations learned by EfficientNet to the number of classes in the leaf disease dataset. Dropout layers are incorporated to prevent overfitting by randomly deactivating certain neurons during training, ensuring that the model does not become overly dependent on specific features in the training data, thus improving generalization on unseen test data. The model is trained to minimize cross-entropy loss (L), which measures the dissimilarity between the predicted probabilities and the actual class labels. The formula for cross-entropy loss is:

$$L(p, y) = - \sum_{i=1}^C y_i \log(p_i) \quad (2)$$

where p_i is the predicted probability for class i , y_i is the binary indicator (0 or 1) if class i is the correct class, and C is the total number of classes. This loss function effectively penalizes incorrect classifications while rewarding correct predictions, helping the model to learn how to better differentiate between leaf diseases.

The model is optimized using stochastic gradient descent (SGD) with momentum, which helps accelerate gradient descent in the right direction, leading to faster convergence. Training is performed for 50 epochs with a batch size of 32, using a learning rate of 0.001 (adjusted with a cosine annealing schedule). A dropout rate of 0.3 is applied in fully connected layers to prevent overfitting, while weight decay ($1e-4$) ensures better generalization. Additionally, early stopping is employed during training, which halts the process when the validation accuracy plateaus or begins to degrade.

C. Real-Time Prediction with FastAPI

For practical deployment, the trained model is integrated with FastAPI, a modern, high-performance web framework designed for quick, real-time prediction services. FastAPI is lightweight, asynchronous, and allows seamless interaction between users, such as farmers or agronomists, and the trained model via HTTP requests. The prediction pipeline consists of several key steps. First, FastAPI accepts an image input from the user, typically through a mobile app or web interface, where the image is uploaded via an HTTP POST request in the appropriate format (JPEG, PNG). Next, the received image is resized and normalized using the same techniques applied during model training, ensuring consistency between training and inference phases. The preprocessed image is then passed through the EfficientNet model, which outputs a probability distribution over the leaf disease classes. FastAPI returns a JSON response containing the predicted class label and confidence scores for each class. This enables users to make informed decisions about the health of the plant and take appropriate actions. FastAPI's scalability allows the system to handle multiple requests simultaneously, ensuring that it can be deployed in real-world environments.

D. Precautionary tips and Recommendations

After the classification of the leaf as healthy or diseased, the system provides precautionary tips and treatment recommendations based on the detected disease. These tips are tailored to help farmers take preventive or corrective actions promptly, minimizing crop loss and preventing the spread of the disease. For example, if a fungal infection is detected, the system might suggest applying specific fungicides or adjusting irrigation practices to reduce moisture levels. In cases of bacterial infections, it may recommend removing affected plants to contain the spread. Additionally, the system can offer general agricultural advice, such as monitoring environmental conditions, improving soil health, or adopting crop rotation practices to prevent future outbreaks. These recommendations are provided along with the prediction results, ensuring that farmers not only receive accurate diagnostics but also practical guidance to address the issues effectively. By integrating disease detection with actionable tips, the system empowers farmers to make informed decisions and enhance crop productivity.

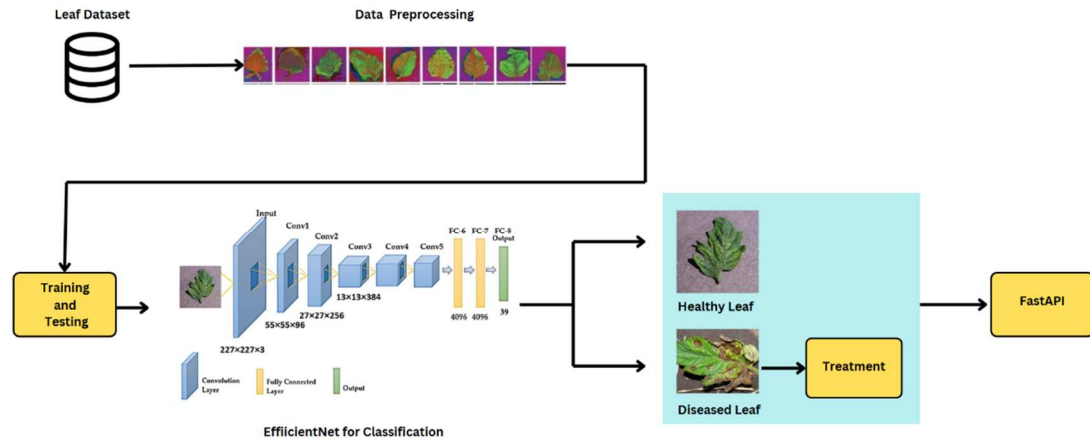


Figure 1 System Architecture

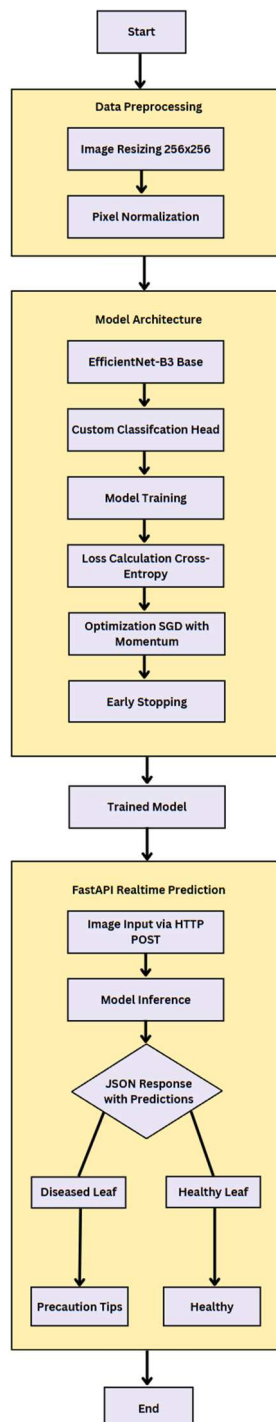


Figure 2 illustrates the proposed methodology of the system, which integrates data preprocessing, model training, and real-time prediction. The process begins with image resizing to 256x256 pixels and pixel normalization to prepare the input data. The model leverages the EfficientNet-B3 architecture with a custom classification head, optimized using cross-entropy loss and Stochastic Gradient Descent (SGD) with momentum. Early stopping ensures optimal training by preventing overfitting. The trained model is deployed using FastAPI, enabling real-time predictions through HTTP POST requests. During inference, incoming images undergo preprocessing, and the model returns predictions in JSON format, classifying leaves as healthy or diseased for appropriate treatment recommendations.

IV. SYSTEM ARCHITECTURE

The system architecture in Figure 1 demonstrates a leaf disease detection pipeline initiated with a tomato leaf dataset. The preprocessing module implements critical transformations: image resizing to 227x227x3 dimensions, normalization to [0,1] scale, and augmentation techniques including rotation and flipping to enhance model generalization [5, 6].

The core classification engine employs EfficientNet with five convolutional layers (Conv1-Conv5) followed by the three fully connected layers (FC-6, FC-7, FC-8). The network progressively reduces spatial dimensions from 227x227x3 to 55x55x96, 27x27x256, and 13x13x384 through convolutional stages, while increasing feature depth. The fully connected layers compress these features to 4096 neurons before the final 39-class output layer [9, 10]. This architecture implements squeeze-and-excitation blocks for adaptive feature recalibration.

The deployment infrastructure utilizes FastAPI, implementing RESTful endpoints for real-time inference with asynchronous processing capabilities. Upon detection of diseased leaves, the treatment module activates, accessing a knowledge base to provide disease-specific protocols [7, 8]. The system maintains a modular microservices architecture, with containerized components ensuring scalable deployment and seamless integration [12, 13].

V. RESULT AND ANALYSIS

The proposed model achieves outstanding performance across ten tomato leaf categories, including bacterial, fungal, viral infections, and healthy plants. With precision, recall, and F1-scores close to 1.00 for all categories, the model demonstrates near-perfect classification. This indicates the system's ability to minimize both false positives and false negatives effectively.

The EfficientNet-B3 architecture ensures optimal feature extraction with fewer parameters, enabling faster convergence and efficient training. The model stabilizes after 10 epochs and 10,077 images, with validation accuracy peaking at 99.4%, showing its robustness on unseen data. Due to its lightweight design and GPU acceleration, the system supports real-time predictions with minimal resource requirements, making it highly suitable for deployment in precision agriculture to ensure timely disease management.

To further validate the model's generalization ability, 5-fold cross-validation was performed. The dataset was split into five equal parts, training on four folds while testing on the remaining one, rotating this process across all folds. The model achieved an average validation accuracy of 98.9% with a standard deviation of 0.3%, confirming its robustness across different data splits. This approach minimizes overfitting risks and ensures consistent performance on unseen data.

Table II presents the performance metrics with Precision, Recall, and F1-Score for each class in the tomato leaf disease classification model. The results demonstrate the model's exceptional ability to identify and classify both diseased and healthy leaves with high accuracy. Most diseases, including Tomato Bacterial Spot, Tomato Early Blight, and Tomato Yellow Leaf Curl Virus, achieve perfect scores across all metrics, reflecting the model's precision in distinguishing these diseases. A slight drop in recall for Tomato Septoria Leaf Spot (0.98) suggests a small number of false negatives, but overall, the high F1-scores indicate balanced precision and recall across all classes. These metrics highlight the effectiveness of the EfficientNet architecture and the adequacy of the dataset used, ensuring confident predictions to aid farmers.

TABLE II Performance Metrics

S. No	Name of the Disease	Precision	Recall	F1-Score	Accuracy
1	Tomato_Bacterial_Spot	1.00	1.00	1.00	99.0%
2	Tomato_Early_Blight	0.99	1.00	0.99	98.9%
3	Tomato_Late_Blight	0.99	1.00	1.00	99.1%
4	Tomato_Leaf_Mold	1.00	1.00	1.00	99.0%
5	Tomato_Septoria_Leaf_Spot	1.00	0.98	0.99	98.7%
6	Tomato_Two-spotted_Spider_Mites	1.00	1.00	1.00	99.0%
7	Tomato_Target_Spot	1.00	1.00	1.00	99.3%
8	Tomato_Yellow_Leaf_Curl_Virus	1.00	1.00	1.00	99.1%
9	Tomato_Mosaic_Virus	1.00	1.00	1.00	99.2%
10	Tomato_Healthy	1.00	1.00	1.00	99.5%

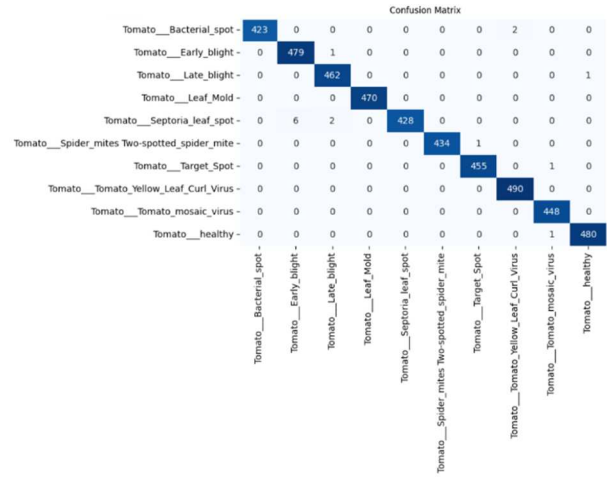


Figure 3 Plots of confusion matrix

Figure 3 presents the confusion matrix illustrating the performance of the classification model on different tomato leaf diseases. Each row in the matrix represents the actual class, while each column corresponds to the predicted class, with the diagonal values indicating correct predictions. The matrix highlights that the model performs exceptionally well, achieving high accuracy across all tomato-related diseases, including Tomato Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot, Target Spot, Yellow Leaf Curl Virus, and Mosaic Virus, along with Healthy Tomato leaves. The low number of misclassifications between these categories confirms the model's precision and ability to distinguish between visually similar diseases. This strong performance, demonstrated by the concentration of values along the diagonal, reflects the effectiveness of the EfficientNet architecture in processing complex leaf patterns, ensuring reliable disease detection for real-world agricultural applications.

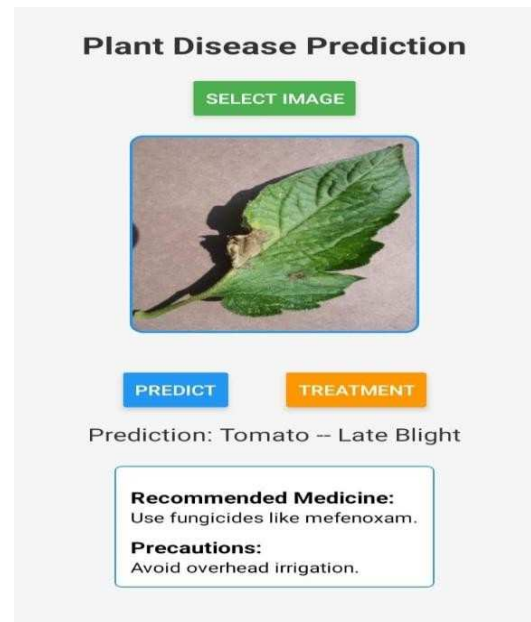


Figure 4 Mobile App for Disease Prediction and Precaution Tips

To leverage the capabilities of the EfficientNet-B3 algorithm for practical use, we developed a mobile application using React Native. The app provides an intuitive interface where users, such as farmers or agronomists, can upload images of plant leaves to diagnose diseases in real time. As shown in the figure 4, after uploading an image and receiving the classification result (e.g., Tomato – Late Blight), the app offers recommended treatment tips along with precautionary tips.

This combination of predictions with actionable guidance, such as the use of fungicides like mefenoxam and avoiding overhead irrigation, ensures users can make informed decisions to protect their crops. The integration with the backend FastAPI service enables seamless, fast predictions, making the system scalable for real-world agricultural applications.

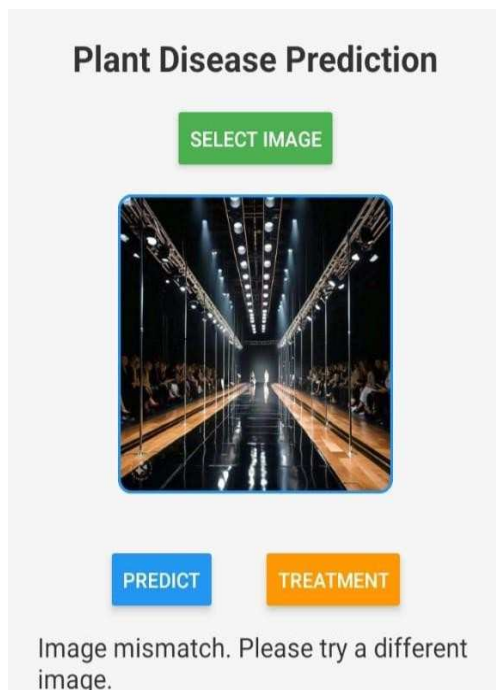


Figure 5 Error Handling for Image Mismatch in Mobile App

To enhance the usability and reliability of the system, the mobile app incorporates error handling mechanisms for invalid inputs. As shown in the figure 5, when an image that is not related to plant leaves (e.g., a fashion runway) is uploaded, the system detects the mismatch and returns a prompt: "Image mismatch. Please try a different image." This feature ensures that the model processes only valid inputs, reducing unnecessary computation and preventing incorrect predictions. The validation mechanism improves user experience by guiding them to upload appropriate images, ensuring the system focuses solely on tomato leaf disease classification.

V. CONCLUSION AND FUTURE WORK

This study presents a robust solution for leaf disease classification using the EfficientNet-B3 model, enhanced by data augmentation, normalization, and deployed through FastAPI for real-time predictions. The system achieves high accuracy and scalability, offering practical benefits for agricultural stakeholders through fast and reliable disease identification. Future work will focus on increasing the dataset's diversity to cover more plant species and disease variations, as well as experimenting with lightweight architectures for seamless deployment on mobile and edge devices to enable offline usage. Exploring self-learning mechanisms through user feedback and incremental model updates could enhance adaptability over time. Additionally, integrating multi-modal data such as environmental factors (e.g., weather, soil quality) with visual analysis may further refine predictions, paving the way for a more intelligent, context-aware crop management system.

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